

# LABORATORY OBSERVATION/RECORD

**Course:** Full Stack Web Development

**Course Code:** 23CM4121

**Experiment No.:** 1

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**Roll No.:** A24126552068

**Date:** 31-08-2026

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## 1. TITLE

Design of a College Information Portal Webpage Using HTML

## 2. AIM

To design a static web page using HTML that demonstrates text formatting, ordered and unordered lists, hyperlinks, tables, images, a registration form with various input types, and embedded audio/video elements.

## 3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML document.
- To use semantic tags such as header, nav, section, and footer.
- To apply text formatting tags for bold, italic, underline, subscript and superscript text.
- To create ordered and unordered lists.
- To insert hyperlinks that open in a new browser tab.
- To display tabular data using table, tr, th and td tags.
- To embed images with appropriate alt text and sizing.
- To design a registration form using various input types (text, email, password, date, number, color, range, radio, checkbox, select, textarea).
- To embed audio and video content with playback controls.

## 4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create the structure and content of web pages. An HTML document is composed of nested elements, each represented by a start tag and an end tag.

Semantic tags such as <header>, <nav>, <section>, and <footer> describe the meaning of the content they contain, making the page easier to read and maintain than using generic <div> tags everywhere.

Text formatting tags (<b>, <i>, <u>, <sub>, <sup>) control the visual presentation of text. Lists (<ol> and <ul>) organize related items, while the anchor tag (<a>) creates hyperlinks to other pages or resources.

Tables (<table>, <tr>, <th>, <td>) arrange data into rows and columns, and the <img> tag embeds pictures using a source path, alternate text, and display dimensions.

Forms (<form>) collect user input through a variety of input types, dropdowns, and text areas. The <audio> and <video> tags embed media that can be played directly in the browser using built-in controls.

## 5. ALGORITHM / PROCEDURE

1. Create a new HTML file with the .html extension.
2. Set up the basic document structure using <!DOCTYPE html>, <html>, <head> and <body> tags.
3. Add a <header> section with the portal title and a welcome message.
4. Add a <nav> section with links to About, Students, Registration and Media sections.
5. Create an About section demonstrating text formatting, ordered/unordered lists, and hyperlinks.
6. Create a Students section containing a table of student details and sample images.
7. Create a Registration section with a form containing various input types.
8. Create a Media section embedding an audio player and a video player.
9. Add a <footer> with copyright information.
10. Save the file and open it in a web browser.
11. Verify that each section renders as expected.
12. Test navigation links, form inputs, and media playback.
13. Record the observed output and compare it with the expected output.

## 6. PROGRAM

### index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>

<body>

<header>
  <h1>College Information Portal</h1>
  <p>Welcome to the College Information Portal.</p>
</header>

<nav>
  <a href="#about">About</a> |
  <a href="#students">Students</a> |
  <a href="#registration">Registration</a> |
  <a href="#media">Media</a>
</nav>

<section id="about">

  <article>

    <h2>About Our College</h2>

    <p>
```

## LABORATORY OBSERVATION/RECORD

ANITS is one of the leading engineering colleges offering quality education.

</p>

<h3>Text Formatting</h3>

<p>

<b>Bold Text</b><br>

<i>Italic Text</i><br>

<u>Underlined Text</u><br>

H<sub>2</sub>O (Subscript)<br>

x<sup>2</sup> (Superscript)

</p>

<h3>Ordered List</h3>

<ol>

<li>CSE</li>

<li>AIML</li>

<li>ECE</li>

<li>EEE</li>

</ol>

<h3>Unordered List</h3>

<ul>

<li>Library</li>

<li>Laboratories</li>

<li>Sports</li>

<li>Hostel</li>

</ul>

<h3>Hyperlinks</h3>

<a href="https://anits.edu.in" target="\_blank">

College Website

</a>

<br><br>

<a href="https://www.google.com" target="\_blank">

Google

</a>

</article>

</section>

<section id="students">

<h2>Student Details</h2>

<table border="1">

<tr>

<th>Roll No</th>

<th>Name</th>

<th>Branch</th>

<th>Email</th>

</tr>

## LABORATORY OBSERVATION/RECORD

```
<tr>
  <td>101</td>
  <td>Bhanu</td>
  <td>AIML</td>
  <td>bhanu@gmail.com</td>
</tr>

<tr>
  <td>102</td>
  <td>Rahul</td>
  <td>CSE</td>
  <td>rahul@gmail.com</td>
</tr>

<tr>
  <td>103</td>
  <td>Priya</td>
  <td>ECE</td>
  <td>priya@gmail.com</td>
</tr>
</table>

<h2>Images</h2>




</section>

<section id="registration">
  <h2>Student Registration Form</h2>
  <form>
    <label>Name:</label>
    <input type="text" placeholder="Enter Name">

    <br><br>

    <label>Email:</label>
    <input type="email" placeholder="Enter Email">

    <br><br>

    <label>Password:</label>
    <input type="password">

    <br><br>

    <label>Date of Birth:</label>
    <input type="date">

    <br><br>

    <label>Age:</label>
```

## LABORATORY OBSERVATION/RECORD

```
<input type="number">

<br><br>

<label>Favourite Color:</label>
<input type="color">

<br><br>

<label>Skill Level:</label>
<input type="range" min="1" max="10">

<br><br>

<label>Gender:</label>
<br>

<input type="radio" name="gender"> Male
<input type="radio" name="gender"> Female
<input type="radio" name="gender"> Other

<br><br>

<label>Hobbies:</label>
<br>

<input type="checkbox"> Reading
<input type="checkbox"> Sports
<input type="checkbox"> Music

<br><br>

<label>Branch:</label>

<select>
  <option>CSE</option>
  <option>AIML</option>
  <option>ECE</option>
  <option>EEE</option>
</select>

<br><br>

<label>Comments:</label>

<br>

<textarea rows="5" cols="40"></textarea>

<br><br>

<input type="submit" value="Submit">
<input type="reset" value="Reset">

</form>

</section>

<section id="media">
```

## LABORATORY OBSERVATION/RECORD

```
<h2>Audio</h2>

<audio controls>
  <source src="college.mp3" type="audio/mpeg">
  Your browser does not support audio.
</audio>

<h2>Video</h2>

<video width="400" controls>
  <source src="college.mp4" type="video/mp4">
  Your browser does not support video.
</video>

</section>

<footer>

  <p>
    &copy; 2026 College Information Portal. All Rights Reserved.
  </p>

</footer>

</body>
</html>
```

## 7. CODE EXPLANATION

| Code                                                                                      | Explanation                                                                |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <html>, <head>, <body>                                                                    | Defines the overall structure of the web page.                             |
| <header>, <nav>, <section>, <footer>                                                      | Semantic HTML5 tags used to organize content into logical regions.         |
| <b>, <i>, <u>, <sub>, <sup>                                                               | Used to bold, italicize, underline, and format subscript/superscript text. |
| <ol>, <ul>, <li>                                                                          | Create ordered and unordered lists of items.                               |
| <a href target='_blank'>                                                                  | Creates a hyperlink that opens the destination in a new tab.               |
| <table>, <tr>, <th>, <td>                                                                 | Displays student details in a tabular format.                              |
| <img src alt width>                                                                       | Embeds images with fallback alt text and a fixed display width.            |
| <form>                                                                                    | Container for all input controls used to collect user data.                |
| <input type='text'/'email'/'password'/'date'/'number'/'color'/'range'/'radio'/'checkbox'> | Different input types used to capture varied kinds of user data.           |
| <select>, <option>                                                                        | Provides a dropdown list for selecting the branch.                         |
| <textarea>                                                                                | Allows multi-line text input for comments.                                 |
| <audio controls>, <video controls>                                                        | Embed playable audio and video content with built-in browser               |

## LABORATORY OBSERVATION/RECORD

|  |           |
|--|-----------|
|  | controls. |
|--|-----------|

## 8. TEST CASES AND EXECUTION DATA

The following test cases were executed in a web browser and the results recorded:

| Test Case | Feature           | Input / Action                                        | Expected Result                                                                                    | Actual Result                        | Status |
|-----------|-------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------|--------|
| TC-01     | Page load         | Open the HTML file in a browser                       | All sections (About, Students, Registration, Media) render without errors                          | Page rendered correctly              | Pass   |
| TC-02     | Text formatting   | Check bold, italic, underline, subscript, superscript | Text displays in the correct format (e.g., H <sub>2</sub> O with 2 as subscript)                   | Displayed correctly                  | Pass   |
| TC-03     | Navigation links  | Click About / Students / Registration / Media links   | Page scrolls to the corresponding section                                                          | Scrolled to correct section          | Pass   |
| TC-04     | Student table     | Inspect the student details table                     | Table shows Roll No, Name, Branch, and Email in rows/columns                                       | Table displayed correctly            | Pass   |
| TC-05     | Registration form | Fill in all fields and select options                 | Form accepts text, email, password, date, number, color, range, radio, checkbox and dropdown input | All inputs accepted values correctly | Pass   |

### Additional Validation Tests

| Test Case | Input                              | Expected Result                                                   |
|-----------|------------------------------------|-------------------------------------------------------------------|
| TC-06     | Email field left in invalid format | Browser should flag the email field as invalid on submit          |
| TC-07     | Hyperlinks clicked                 | College Website and Google links should open in a new browser tab |

## 9. OUTPUT

### Output 1: About Section

*Text formatting samples, ordered and unordered lists, and hyperlinks rendered in the browser.*

## LABORATORY OBSERVATION/RECORD

### About Our College

ANITS is one of the leading engineering colleges offering quality education.

#### Text Formatting

##### Bold Text

*Italic Text*

Underlined Text

H<sub>2</sub>O (Subscript)

x<sup>2</sup> (Superscript)

#### Ordered List

1. CSE
2. AIML
3. ECE
4. EEE

#### Unordered List

- Library
- Laboratories
- Sports
- Hostel

#### Hyperlinks

[College Website](#)

[Google](#)

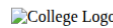
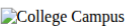
### Output 2: Students Section

*Student details table (image placeholders shown as broken icons since local image files were not supplied).*

#### Student Details

| Roll No | Name  | Branch | Email           |
|---------|-------|--------|-----------------|
| 101     | Bhanu | AIML   | bhanu@gmail.com |
| 102     | Rahul | CSE    | rahul@gmail.com |
| 103     | Priya | ECE    | priya@gmail.com |

#### Images

### Output 3: Registration Form

*Registration form filled with sample data across all input types (text, email, password, date, number, color, range, radio, checkbox, select, textarea).*



## LABORATORY OBSERVATION/RECORD

### Student Registration Form

Name:

Email:

Password:

Date of Birth:

Age:

Favourite Color:

Skill Level:

Gender:  
☒ Male ☐ Female ☐ Other

Hobbies:  
☒ Reading ☒ Sports ☐ Music

Branch:

Comments:  

This form demonstrates various HTML5 input types.

### Output 4: Media Section

*Audio and video players rendered with native browser controls.*

#### Audio



#### Video



## 10. RESULT

Thus, a College Information Portal web page was successfully designed using HTML. Text formatting, lists, hyperlinks, tables, images, a multi-input registration form, and embedded audio/video elements were implemented and verified through multiple test cases.